Sahaj Bhadja

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Summary

Data scientist-minded engineer with 2+ years building **experimentation-ready metrics**, **SQL/Python analysis pipelines**, and **automated reporting workflows** to drive product prioritization and member-experience decisions.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **product-facing metrics dashboards** for 500+ users, reducing SQL latency from **1.2s to 350ms** and enabling more frequent, trustworthy decision cycles for feature prioritization.
- Implemented **data quality validation pipelines** across 5K+ monthly jobs, improving experiment-readiness of datasets and confidence in trend interpretation for cross-functional product and engineering stakeholders.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Developed **Python analysis pipelines** over 100K+ trajectory logs, applying descriptive and inferential methods to quantify behavior shifts and uncover statistically meaningful performance differences.
- Generated rigorous experiment summaries and visual diagnostics, accelerating hypothesis evaluation and helping teams translate technical findings into product-level recommendations with clear uncertainty communication.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Analyzed usage behavior from 10K+ daily requests while shipping features, framing **testable hypotheses** around user friction points and guiding iterative improvements to grading workflows.
- Improved experiment iteration speed with CI/CD controls and concurrent processing, reducing turnaround time by **60%** and increasing cadence for feature evaluation and follow-up analysis.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Built **API and ETL pipelines** using OOP, AWS Glue, and S3, improving response time by **30%** and strengthening reliability of downstream analytical consumption.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, boosting pipeline efficiency by **68%** and reducing data-noise from preventable production regressions.

LST Capitals **India**
Web Developer Intern *November 2022 - January 2023*
- Instrumented **workflow event capture** across 50K+ records in booking and payment flows, improving funnel visibility and enabling better diagnosis of user drop-off stages.
- Built webhook-triggered reporting hooks and documentation, improving stakeholder access to operational insights and shortening turnaround from issue identification to prioritized product action.

Skills

Analytics & Stats: **A/B testing**, quasi-experiment framing, descriptive/inferential statistics, metric design
Languages: **SQL**, **Python**, JavaScript/TypeScript, Java
Data Science: hypothesis generation, cohort/funnel analysis, behavioral segmentation, model-aware interpretation
Reporting: dashboard development, rollout/monitoring tables, automated reporting workflows, stakeholder storytelling
Tooling: Git, CI/CD, Flask/FastAPI/Node.js, PostgreSQL/MySQL/MongoDB, collaborative Agile delivery

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built a **self-serve analytics workspace** across 10+ databases with schema-aware query generation, enabling rapid hypothesis checks and reproducible exploration of product and user-behavior questions.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Implemented **metrics instrumentation** for workflows serving 50K+ users and 1M+ transactions, supporting funnel analysis and data-backed prioritization of retention and conversion improvements.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **operational metric tracking** across 100+ daily appointment workflows, improving record retrieval by **2x** and enabling clearer attribution of performance gains to workflow changes.